The approximate sampling distribution of the test statistic $Z = \dfrac{(\bar{X} - \bar{Y})}{\sqrt{\dfrac{\sigma_X^2}{m} + \dfrac{\sigma_Y^2}{n}}}$ under H_0 (*i.e.*, $\mu_X = \mu_Y$) is the *N(0,1)* distribution.

It may be noted that the test statistic, when $\sigma_X^2 = \sigma_Y^2 = \sigma^2$, is $Z = \dfrac{(\bar{X} - \bar{Y})}{\sigma\sqrt{\dfrac{1}{m} + \dfrac{1}{n}}}$.

Step 5 : Calculate the value of Z for the given samples $(x_1, x_2, ..., x_m)$ and $(y_1, y_2, ..., y_n)$ as $z_o = \dfrac{(\bar{x} - \bar{y})}{\sqrt{\dfrac{\sigma_X^2}{m} + \dfrac{\sigma_Y^2}{n}}}$.

Here, $\bar{x}$ and $\bar{y}$ are respectively the values of $\bar{X}$ and $\bar{Y}$ for the given samples.

Step 6 : Find the critical value, z_e, corresponding to α and H_1 from the following table

Alternative Hypothesis (H_1)	$\mu_X \neq \mu_Y$	$\mu_X > \mu_Y$	$\mu_X < \mu_Y$
Critical Value (z_e)	$z_{\alpha/2}$	z_α	$-z_\alpha$

Step 7 : Make decision on H_0 choosing the suitable rejection rule from the following table corresponding to H_1.

Alternative Hypothesis (H_1)	$\mu_X \neq \mu_Y$	$\mu_X > \mu_Y$	$\mu_X < \mu_Y$
Rejection Rule	$\lvert z_0 \rvert \geq z_{\alpha/2}$	$z_0 > z_\alpha$	$z_0 < -z_\alpha$

Example 1.10

Performance of students of X Standard in a national level talent search examination was studied. The scores secured by randomly selected students from two districts, *viz.*, D_1 and D_2 of a State were analyzed. The number of students randomly selected from D_1 and D_2 are respectively 500 and 800. Average scores secured by the students selected from D_1 and D_2 are respectively 58 and 57. Can the samples be regarded as drawn from the identical populations having common standard deviation 2? Test at 5% level of significance.

Solution:

Step 1 : Let μ_X and μ_Y be respectively the mean scores secured in the national level talent search examination by all the students from the districts D_1 and D_2 considered for the study. It is given that the populations of the scores of the students of these districts have the common standard deviation $\sigma = 2$. The null and alternative hypotheses are

Null hypothesis: H_0: $\mu_X = \mu_Y$

i.e., average scores secured by the students from the study districts are not significantly different.

Alternative hypothesis: H_1: $\mu_X \neq \mu_Y$

i.e., average scores secured by the students from the study districts are significantly different. It is a two-sided alternative.

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